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[23EC34]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

MICROCONTROLLER AND ITS APPLICATIONS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 X 2=10**
1.	a.	What role does RAM play in computer systems?	L2	C01	02
	b.	Describe the role of TRIS Register of PIC microcontroller.	L2	C02	02
	c.	State whether the following instruction is valid or invalid: i. MOVWL 40H ii. MOVLW 40H	L2	C03	02
	d.	Define interrupt.	L2	C04	02
	e.	What is the use of Timers?	L2	C05	02
		PART-B			**6 X 5=30**
2.	a.	Differentiate between embedded processors and microcontrollers.	L2	C01	05
		OR			
	b.	Distinguish between Harvard and Von Neumann architectures.	L2	C01	05
3.	a.	With a neat diagram, explain the internal organization of the computer.	L2	C01	05
		OR			
	b.	List the three types of buses found in computer systems and explain briefly the purpose of each type of bus.	L2	C02	05
4.	a.	Explain the following blocks of PIC Microcontroller: i. WREG(1M) ii. ALU(2M) iii. Program Counter(2M)	L2	C02	05
		OR			
	b.	Write a C program to convert data from BCD "0x46" to its ASCII values of PIC microcontroller.	L3	C03	05
5.	a.	Write a C program to convert data from ASCII values "0x34" and "0x36" to its BCD value of PIC microcontroller.	L3	C03	05
		OR			
	b.	Write a C program to convert data from Hexa value "0xFD" to its decimal value of PIC microcontroller.	L2	C03	05

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6.	a.	Calculate TMR0H and TMR0L, for the time delay of 10 ms for crystal frequency of 10 MHz, select no prescalar.	L2	C04	05
		OR			
	b.	Find the value for T0CON, if we want to program Timer0 in 16-bit mode, no prescaler. Use PIC18's Fosc/4 crystal oscillator for the clock source.	L2	C04	05
7	a.	Describe the following communication: i. Simplex Mode ii. Half Duplex iii. Full Duplex	L2	C05	05
		OR			
	b.	With XTAL = 10MHz, find the SPBRG value needed to have the following baud rates: i. 9600 ii. 2400	L3	C05	05
		PART-C		**6 X10=60**	
8	a.	With neat diagram, describe the CPU's instruction execution cycle in detail.	L2	C01	10
		OR			
	b.	Explain the following arithmetic and logical instructions with suitable examples. i. ADDLW K ii. IORLW K iii. SUBLW K iv. XORLW K v. MULLW K	L2	C01	10
9.	a.	Briefly explain different unconditional and conditional jump instructions of PIC Microcontroller with suitable examples.	L3	C02	10
		OR			
	b.	Analyze different C data types supported by PIC Microcontroller.	L2	C02	10
10	a.	Define assembler directives. With example explain all the assembler directives supported by PIC microcontroller.	L2	C02	05
	b.	Show the status of the C, DC, and Z flags after the addition of 38H and 2FH.	L4	C02	05
		OR			

	a.	Write a C program to toggle 8 LEDs continuously with some delay; LEDs are connected to Port B of PIC microcontroller.	L3	C03	05
	b.	Write a C program to get a byte of data from Port B, wait for 500 milli second, and then send it to Port C.	L3	C03	05
11.	a.	With neat functional block diagram, describe the architecture features of PIC microcontroller.	L2	C03	10
		OR			
	b.	Explain the bit contents of TOCON, TMR0H and TMR0L registers of Timer 0.	L4	C04	10
12.	a.	**i.** Compare between assembly language over C language. **ii.** Compare between Serial communication over Parallel communication.	L2	C04	10
		OR			
	b.	Compare polling and interrupts. What are the steps a microcontroller performs upon activation of Interrupt?	L2	C05	10
13.	a.	Draw a 16x2 LCD pin diagram and describe each pin in detail. Also give commands to configure the 16X2 LCD.	L3	C05	10
		OR			
	b.	Write a C program to monitor the status of switch(SW) which is connected to RB0 and stepper motor is connected to PORTC of PIC microcontroller, perform the following: **i.** If SW = 0, move stepper motor clockwise. **ii.** If SW = 1, move stepper motor anticlockwise. Also draw an interface diagram.	L3	C05	10